**Princess Nourah bint Abdulrahman University**
College of Computer and Information Sciences

# White board
# Learning management systems

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## Introduction

In many academic institutions, existing digital platforms have played a central role in supporting online and blended learning. These systems typically offer access to course materials, assignments, announcements, and communication tools that facilitate interaction between students and instructors. Despite their widespread adoption, several persistent challenges remain, most notably, the delay in delivering critical notifications to students.

Students often receive alerts about new assignments, exams, or instructor messages after key deadlines have passed. This delay disrupts academic planning, reduces student responsiveness, and can negatively impact performance. In fast-paced academic environments, timely access to information is essential for managing workload, meeting deadlines, and maintaining engagement.

To address these limitations, this project introduces a standalone academic notification system that operates independently from existing platforms. The system is designed to solve the same core problems—such as delayed alerts and lack of customization—while offering enhanced solutions. It delivers real-time push notifications when new assignments or exams are posted and when grades are released. Notifications are personalized based on user roles (students, instructors, administrators) and academic calendars, ensuring relevance and timeliness Beyond solving known issues, the system introduces additional improvements. These include multi-channel delivery (mobile, email, dashboard), intelligent scheduling, and role-based customization. The system does not replicate or integrate with existing platforms, but instead reimagines the notification experience with a student-centered Approach.

Ultimately, this project reflects a strategic commitment to improving the quality of digital education. By ensuring that academic alerts are timely, relevant, and accessible, the system empowers students to stay informed and proactive, while supporting instructors and administrators in streamlining communication and reducing missed deadlines [1].

## Problem statement

- Traditional education systems faced major challenges such as limited difficulties in submitting and communication between instructors and students and limited access to, lack of centralized course management, tracking assignments learning materials outside the classroom [2].

- E-learning and learning management systems (LMSs) were developed to address these challenges by providing digital environments that allow interaction, online assessments, and easy access to educational content [2].

- However, despite their success in improving digital learning, many existing e-learning systems still face technical and usability issues that affect efficiency,

engagement, and overall user experience[3].

- Common issues in current LMSs include:
- Delayed notifications, where alerts and updates are not delivered instantly, leading to missed announcements or deadlines [4].
- Slow system performance during high-demand periods, such as examinations or assignment submissions [3].
- Synchronization problems between web and mobile versions, causing inconsistent or missing data [4].
- Outdated or complex user interfaces that reduce accessibility and make navigation difficult [3]
- Limited feedback and progress tracking tools for both students and instructors [3].

- These limitations negatively impact communication, engagement, and satisfaction among users [3].

- Therefore, this project proposes the development of an independent e-learning system designed to overcome these issues by introducing advanced improvements, including:
- A real-time push notification system that ensures immediate delivery of updates and reminders [4].
- A multi-channel notification feature that delivers alerts via email and SMS to guarantee reliability [4].
- Enhanced synchronization between devices and platforms for consistent data flow [4].
- An interactive and user-friendly interface (UI/UX) to simplify navigation and improve usability [3].
- Improved performance and faster system response to support high user activity efficiently [3].

- By addressing these aspects, the proposed system aims to provide a more efficient, engaging, and reliable learning experience for all users [2] [3][4].

## Methodology with justification

This project adopts the Incremental Model, which is suitable for systems that require gradual development and improvement. Since the main objective is to enhance the Blackboard Learning Management System (LMS) by resolving the delay in notifications, this model allows the system to be built and delivered in smaller functional parts known as increments. Each increment adds new functionality, ensuring that improvements are introduced step by step and tested before full deployment.

Stages of the Incremental Model:

1. Requirements Gathering:

Collect feedback from students and instructors to identify the root causes of delayed notifications and determine user expectations for faster updates.

2. System Analysis:

Examine technical factors such as synchronization issues and data flow between the Blackboard server and the mobile application to understand the source of delays.

3. Design Phase:

Plan and design the architecture of an enhanced notification system that supports real time message delivery and multi-channel alerts.

4. Implementation (in Increments):

- Increment 1: Develop and integrate an improved in-app notification system to reduce internal delivery delays.

- Increment 2: Introduce external push notifications via email and SMS for broader accessibility.

- Increment 3: Conduct user testing and refine the system based on collected feedback.

5. Testing Phase:

Test each increment individually to verify functionality, reliability, and notification delivery speed before moving to the next phase.

6. Evaluation Phase:

Evaluate the complete system performance after all increments are integrated, focusing on overall efficiency, response time, and user satisfaction.

Reasons for Using the Incremental Model

- Continuous Delivery: Each increment delivers a usable feature, allowing early testing and evaluation.
- Flexibility: Enhancements or adjustments can be added in later increments without redesigning the entire system.
- Risk Reduction: Frequent testing after each increment minimizes the risk of systemwide failures.

- User Involvement: Students and instructors can interact with each version and share feedback to guide subsequent development.
- Efficient Resource Use: Developers can concentrate on one component at a time, ensuring higher quality and maintainable progress.

Comparison with Other Models

Unlike the Waterfall model, which requires completing all phases before producing a functional version, the Incremental model supports partial delivery and progressive improvement. It is also more structured and predictable than Agile/Scrum, making it ideal for projects like this one — where the focus is on steady enhancement and reliability testing rather than rapid iteration

## project charter

TABLE 4-1

**Project Title:** Intelligent Academic Notification System

**Date of Authorization:** October 1, 2025

**Project Start Date:** October 1, 2025 **Projected Finish Date:** December 5, 2025

**Key Schedule Milestones:**

- Complete requirement analysis and design by October 20
- Implement and test the improved notification system November 25
- Go-live with the improved notification system by December 5

**Budget Information:** The university has set aside $50,000 for this initiative. Additional funds can be accessed if requi red. The budget will primarily finance software creation, testing instru ments, and cloud notification services

**Project Manager:** Saba, Project Manager Contact: saba@pnu.edu.sa

**Project Objectives:**
1-Enhance the academic platform by providing accurate and timely notifications for key academic events
2-Allow students, instructors, and administrators to customize their notification preferences according to their roles
3-Support multiple notification channels such as email, mobile alerts, and in-system messages to enhance communication
4-Automatically send notifications based on scheduled academic data and calendar updates, without needing real-time synchronization
5-Ensure the notification system operates securely, reliably, and independently from existing academic platforms
6-Improve system speed to guarantee fast delivery of notifications, even during busy academic periods
7-Enable high concurrency to process many notifications simultaneously without delays, conflicts, or duplicates

**Main Project Success Criteria:**
-Notifications are delivered promptly and precisely to each user through their chosen method, such as email, mobile, or dashboard
-Users have the ability to configure their preferences and access their notification records.
-The system operates safely and autonomously, utilizing scheduled data feeds.
-Users are 90% pleased with the system's efficiency and dependability.

**Approach:**

- Carry out surveys among students and faculty to pinpoint existing problems with Blackboard notifications.
- Complete the requirements, scope statement, and detailed schedule within one month
- Create and evaluate new functionalities (real-time push notifications, customizable settings)
- Conduct weekly meetings to review progress with the sponsor and team.
- Conduct comprehensive system testing and initial deployment prior to the complete launch

**ROLES AND RESPONSIBILITIES**

| Name | Role | Position | Contact Information |
|---|---|---|---|
| Saba | Project Manager | Team Member. | saba@pnu.edu.sa |
| Yara | Developer | Team Member | yara@pnu.edu.sa |
| Razan | Designer | Team Member | razan@pnu.edu.sa |
| Budur | programmer | Team Member | Budur@pnu.edu.sa |
| Aryam | programmer | Team Member | aryam@pnu.edu.sa |
| Thekra | programmer | Team Member | thekra@pnu.edu.sa |

**Sign-off: (Signatures of all the above stakeholders)**

| | |
|---|---|
| Saba | Budur |
| Yara | Aryam |
| Razan | Thekra |

**Comments: (Handwritten or typed comments from above stakeholders, if applicable)**

"I am excited to lead this project as a student manager. Improving Blackboard notifications will help students stay more engaged and organized." – Saba

"I'm excited to be part of this project and look forward to helping develop a notification system that truly supports students in keeping up with their tasks and academic deadlines."
— Razan

*Table 1 project charter*

## Techniques for collecting requirements

**1.Stakeholders: Instructors, Academic Administrators**
- **Technique: Interviews**
- **Reason:**

Interviews offer in-depth understanding of academic workflows, notification policies, and approval procedures. They assist in revealing concealed requirements and elucidating regulations that directly affect the system.

**2.Stakeholders: Students**
- **Technique: Questionnaires / Surveys**
- **Reason:**

Surveys effectively collect extensive feedback from students about their preferences for notifications and the issues they face with existing systems. This aids in making knowledgeable choices regarding design and usability.

**3.Stakeholders: IT Support Staff, Project Team**
- **Technique: Focus Groups**
- **Reason:**

Focus groups promote cooperative dialogue among technical personnel and project participants. They assist in pinpointing system issues, improving technical specifications, and confirming the practicality of suggested functionalities.

## Project Requirements

**Software Requirements**
- Reliable authentication and access control based on roles for students, teachers, and administrators.
- Event notification system designed to handle academic occurrences and automatically produce alerts.
- A delivery system that utilizes multiple channels, including email, push notifications, and alerts within the dashboard.

**Hardware Requirements**
- A web server to support the independent notification system
- A database server for storing user information, school calendars, and alert records.
- Network resources adequate to manage peak demands during academic due dates

**Functional Requirements**
- Students should get prompt alerts via their chosen methods (email, dashboard alerts)
- The system should enable users to personalize notification settings and choose to optin or opt-out of particular categories.
- The system should automatically activate alerts according to established academic guidelines.
- The system needs to distinctly differentiate permissions among students, instructors, and administrators

**Non-functional Requirements**
- Scalability: capacity to allocate resources during peak traffic times without compromising performance
- Performance: excellent responsiveness with minimal downtime and rapid data processing
- Security: robust security measures to safeguard user information and adhere to applicable regulations
- Reliability: elevated accessibility and reduced downtime through a sturdy failover management and data recovery system
- Speed: The system needs to provide notifications within seconds of receiving academic events, guaranteeing quick processing and low delay during high activity times
- Concurrency: The system should be capable of managing numerous notification events at the same time, enabling effective processing even during busy acad emic periods without issues, delays, or repetitions

## Scope Statement

# PROJECT SCOPE STATEMENT

Project title: white board learning management system Date Prepared: October 1, 2025

Project Scope Description:

This project aims to develop a standalone academic notification system that intelligently manages and delivers alerts related to students' academic activities. The system addresses limitations observed in existing platforms by introducing enhanced notification mechanisms. Key features .include automatic alerts when new assignments or exams are posted, and when grades are released Notifications are personalized based on user roles and academic schedules. Although the system is not affiliated with any existing academic platform, its design is informed by the challenges and gaps .identified in current systems, and it offers improved solutions independently

Project Deliverables:

- A standalone web-based academic notification system
- Intelligent scheduling engine based on academic calendars and user roles
- Automatic notifications for:
- New assignments or exams posted
- Grade releases for assignments and tests
- Multi-channel delivery: mobile push, email, and dashboard alerts
- Role-based customization for students, instructors, and admins
- Administrative dashboard for managing notification rules and logs
- User interface for managing preferences and viewing notification history
- Technical documentation and training materials

Product acceptance criteria:

- Notifications must be delivered accurately and in a timely manner
- The system must support role-based customization for different user types
- Users must be able to view, manage, and archive their notification history
- The system must operate independently and securely
- All features must comply with institutional privacy and data protection standards

Project exclusions:

- The system does not perform grading, assignment submission, or content delivery itself, but it sends notifications when such events occur (e.g., grade release, new assignment posted)
- The system does not synchronize in real time with institutional databases, but relies on scheduled or manual data inputs to trigger notifications
- It will not replicate or mention the name of any existing academic platform, but will address the same problems and offer improved solutions independently

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## PROJECT SCOPE STATEMENT

Project constraints

- Limited access to institutional academic systems and databases
- Project schedule limitations and fixed deadlines
- Data privacy and compliance requirements
- Potential integration challenges with third-party systems

Project assumptions

- Academic calendars and schedules will be made available to configure the notification engine
- Users (students, instructors, admins) will have access to mobile devices or email to receive notifications
- Feedback from users of current systems will be used to identify gaps and refine features
- The system will be hosted independently and maintained without dependency on institutional infrastructure

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*Table 2 scope statement*

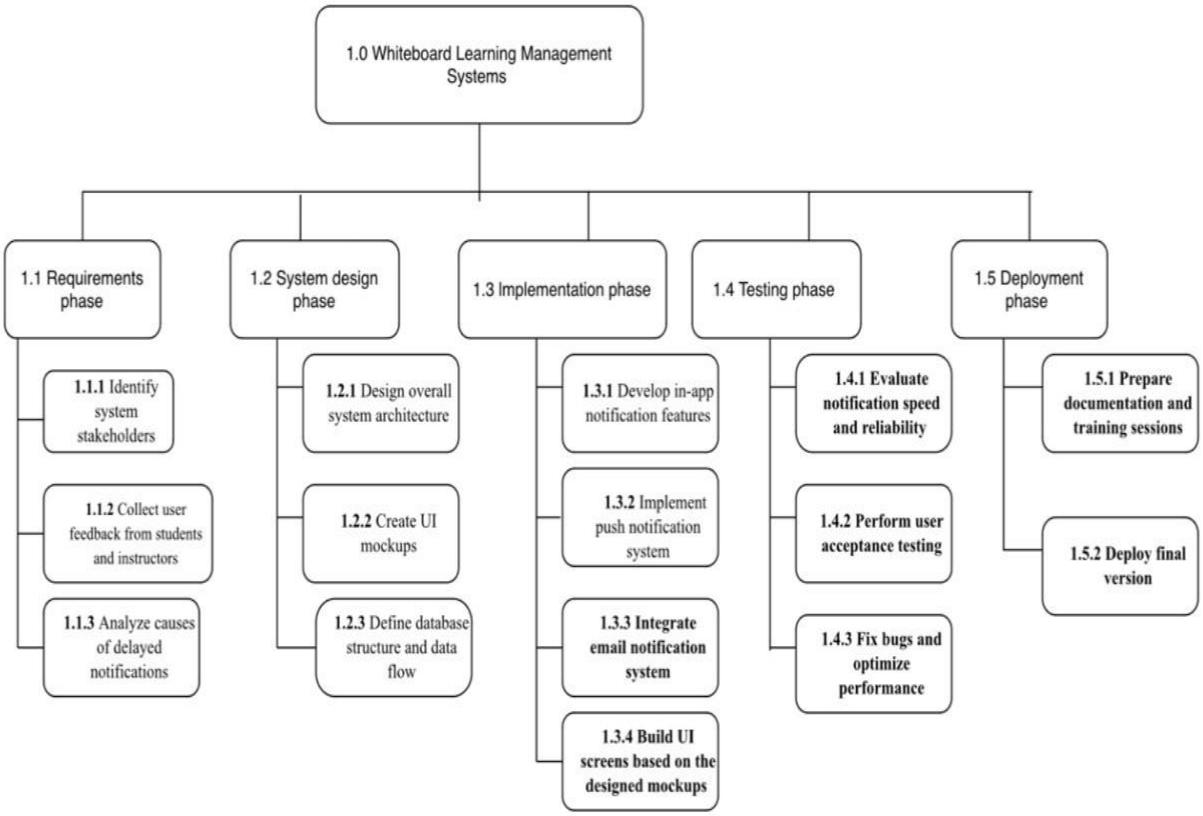



*Figure 1 WBS whiteboard learning management systems project*

## WBS

This figure shows the main phases and tasks of the Whiteboard Learning Management System project. It breaks down the work into five phases: Requirements, Design, Implementation, Testing, and Deployment. Each phase includes specific tasks with IDs to help organize and manage the project more clearly.

## WBS Dictionary- Whiteboard Learning Management Systems

| Task ID | Task Name | Description | Responsible Person | Estimated Duration |
|---|---|---|---|---|
| 1.1.1 | Identify system stakeholders | List and confirm key project stakeholders | Project Manager (PM) | 1 day |
| 1.1.2 | Collect user feedback | Collect user feedback from students and instructors through surveys and interviews | Business Analyst (BA) | 2 days |
| 1.1.3 | Analyze causes of delayed notifications | Identify root causes behind delayed notifications | System Analyst | 2 days |
| 1.2.1 | Design overall system architecture | Design the overall system architecture including modules, interfaces, and integration points | System Architect | 3 days |
| 1.2.2 | Create UI mockups | Create UI prototypes for notification flow | UI/UX Designer | 2 days |
| 1.2.3 | Define database structure and data flow | Define database schema and data flow | Database Engineer | 3 days |
| 1.3.1 | Develop in-app notification features | Implement inside-app notification functionality | Frontend Developer | 4 days |
| 1.3.2 | Implement push notification system | Integrate push notification service | Backend Developer | 4 days |
| 1.3.3 | Integrate email notification system | Develop email alert integration | Backend Developer | 3 days |
| 1.3.4 | Build UI screens | Build UI screens based on the designed mockups | Full-stack Developer | 4 days |
| 1.4.1 | Evaluate notification speed and reliability | Test realtime delivery performance | Tester | 3 days |
| 1.4.2 | Perform user acceptance testing | Conduct UAT with students and instructors | QA Team | 2 days |
| 1.4.3 | Fix bugs and optimize performance | Resolve issues and enhance system performance | Developer | 2 days |
| 1.5.1 | Prepare documentation and training sessions | Create user guides and conduct training | PM / Trainer | 1 days |
| 1.5.2 | Deploy final version | Launch final version to production environment | DevOps Engineer | 1 days |

*Table3 WBS Dictionary - Whiteboard Learning Management Systems*

## WBS dictionary

This table presents the Work Breakdown Structure (WBS) Dictionary for the project. It outlines notification system tasks, including task IDs, names, descriptions, responsible roles, and estimated durations, providing a structured reference for project management and tracking.

## RACI chart

| | PM | BA | System Analyst | System Architect | UI/UX | DB Engineer | Frontend Dev | Backend Dev | Full-Stack Dev | QA/Tester | DevOps |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Identify system stakeholders | R A | C | I | I | I | I | I | I | I | I | I |
| Collect user feedback | A | R | C | I | I | I | I | I | I | I | I |
| Analyze causes of delayed notifications | A | C | R | C | I | C | I | I | I | I | I |
| Design overall system architecture | A | I | C | R | I | C | I | I | I | I | I |
| Create UI mockups | A | I | I | I | R | I | C | C | C | I | I |
| Define database structure & data flow | A | I | C | C | I | R | I | I | I | I | I |
| Develop in-app notification features | A | I | I | I | C | I | R | I | I | I | I |
| Implement push notification system | A | I | I | I | I | I | I | R | I | I | I |
| Integrate email notification system | A | I | I | I | I | I | I | R | I | I | I |
| Build UI screens | A | I | I | I | C | I | C | C | R | I | I |
| Evaluate notification speed & reliability | A | I | C | I | I | I | I | I | I | R | I |
| Perform user acceptance testing | A | C | I | I | I | I | I | I | I | R | I |
| Fix bugs and optimize performance | A | I | I | I | I | I | C | C | R | R | I |
| Prepare documentation & training | R A | I | I | I | I | I | I | I | I | I | I |
| Deploy final version | A | I | I | I | I | I | I | I | I | I | R |

*Table4 RACI matrix assigned to each project task*

## Gantt chart

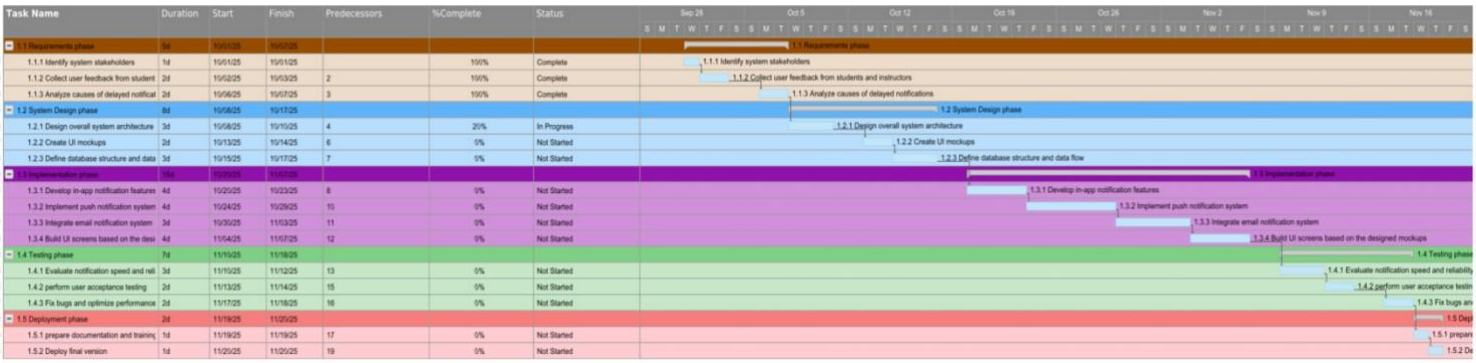



*Figure2 Gantt chart for the Intelligent Academic Notification system project*

Figure 3 presents the Gantt chart for the Intelligent Academic Notification System Project. The chart illustrates the main project phases: requirements gathering, system design, implementation, testing, and final deployment. It also shows the sequence of tasks, their dependencies, and the completion percentage for each activity. This visual representation helps track project progress and understand how tasks flow together until the final system release.

Milestones:

1. Requirements Phase Completed (End of identifying all project requirements)

2. System Design Approved (Architecture and UI/UX design finalized)

3. Development Phase Completed (Core system features implemented)

4. Testing Phase Completed (System successfully tested and validated)

5. Final Project Completed & Submitted (All deliverables done and project officially closed)

## Risk register

| ID | Category | Risk description | Cause | Trigger | Type (Threat/Opportunity) | Probability (1-5) | Impact (1-5) | Score (P×I) | Response strategy | Planned actions | Owner | Due date | Status | Residual risk | Contingency plan |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Schedule | Delay in completing increment due to overlapping tasks | Poor time estimation and task overlap | Sprint tasks spill into next increment | Threat | 3 | 4 | 12 | Mitigate | Refine sprint backlog, allocate buffer days between increments | Project Manager | 2025-11-20 | Open | Medium | Reassign developer or adjust next increment timeline |
| 2 | Requirement | New request from academic staff mid-project | Late feedback from stakeholders | New feature requested after increment delivery | Threat | 3 | 5 | 15 | Avoid | Freeze requirements after initial increment; create change request form | Business Analyst | 2025-11-18 | Open | Low | Defer new requests to later release cycle |
| 3 | Technical | Integration with Blackboard notification API fails intermittently | API limitations or downtime | API response errors or timeouts | Threat | 2 | 5 | 10 | Transfer | Use cached notifications and alert user; contact IT for SLA | Tech Lead | 2025-11-15 | Open | Low | Fallback to manual sync via CSV export |
| 4 | Quality | Bug in notification filter affects accuracy | Insufficient testing in earlier increments | Incorrect notifications displayed | Threat | 3 | 4 | 12 | Mitigate | Add regression tests; peer review code before merge | QA Lead | 2025-11-22 | In progress | Low | Rollback to previous version if critical |
| 5 | Resources | Limited availability of tester during finals week | Tester exam schedule | Testing phase postponed | Threat | 3 | 3 | 9 | Mitigate | Cross-train another member for testing tasks | Project Manager | 2025-11-25 | Open | Medium | Borrow resource from dev team temporarily |
| 6 | Opportunity | University's tech expo aligns with release date | Event timing | Event announcement | Opportunity | 3 | 4 | 12 | Exploit | Prepare demo and marketing material for exposure | PM & Marketing | 2025-11-30 | Planned | Low | Submit demo registration early |
| 7 | Security | User data not encrypted properly in local storage | Improper handling of credentials | Security review findings | Threat | 2 | 5 | 10 | Mitigate | Encrypt sensitive data; add validation before saving | Security Lead | 2025-11-28 | In progress | Low | Disable local caching if issue persists |
| 8 | Infrastructure | Server capacity insufficient for high user load | Limited hosting plan | Slow response time under load | Threat | 2 | 4 | 8 | Mitigate | Upgrade to scalable hosting or use load balancing | DevOps Engineer | 2025-12-01 | Open | Low | Move to backup server if downtime >1h |
| 9 | Opportunity | Mentor from IT department offers technical guidance | Mentor initiative | Mentor contact email | Opportunity | 4 | 3 | 12 | Enhance | Schedule mentoring sessions to improve code quality | Tech Lead | 2025-11-18 | Scheduled | Low | Share lessons learned with full team |
| 10 | Compliance | App not aligned with university IT policy on notification permissions | Lack of policy review | Policy review meeting | Threat | 2 | 4 | 8 | Mitigate | Review policies and update consent text in app | BA / Legal | 2025-11-26 | In progress | Low | Disable push notifications until approval |

*Table5 Risk Register*

| Risk Level | Score Range | Description | Example Risks |
|---|---|---|---|
| High | 15–25 | Requires immediate action and mitigation plan | Stakeholder late requests, API instability |
| Medium | 8–12 | Needs monitoring and controlled response | Testing delay, notification filter bug |
| Low | 1–7 | Acceptable, monitor occasionally | Server load, cost increase |

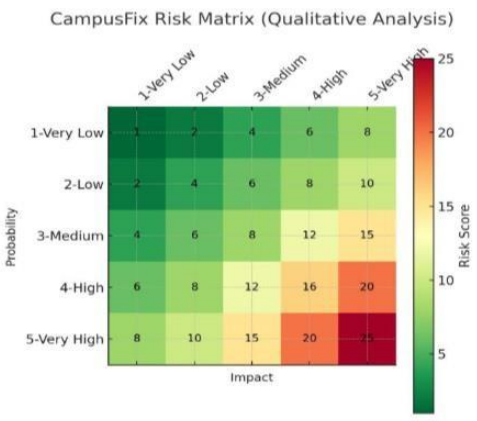



*Figure 3 Risk Matrix*

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